

REPORT

Predicting Modeling Using Machine Learning

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Predictive Modeling Using Machine Learning

Objective

The objective of this project is to build a machine learning model that predicts passenger survival using the Titanic dataset.

Dataset

Titanic Dataset

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Scikit-Learn

Data Preparation

```
[5]: # Fill missing values
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Convert text data into numbers
df["Sex"] = df["Sex"].map({"male": 0, "female": 1})
df["Embarked"] = df["Embarked"].map({"S": 0, "C": 1, "Q": 2})

# Select features for prediction
X = df[["Pclass", "Sex", "Age", "Fare"]]

# Target variable
y = df["Survived"]

print("Data prepared successfully!")

Data prepared successfully!
```

Figure 1: Data Preparation for Machine Learning

Explanation:

The dataset was preprocessed by handling missing values and converting categorical variables into numerical format. The selected features were Passenger Class, Gender, Age, and Fare.

Train-Test Split

```
[6]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data:", len(X_train))
print("Testing data:", len(X_test))
```

Training data: 712

Testing data: 179

Explanation:

The dataset was divided into training and testing sets. Eighty percent of the data was used for training, while twenty percent was reserved for testing model performance.

Model Training

```
[7]: from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)

print("Model trained successfully!")

Model trained successfully!
```

Figure 3: Random Forest Model Training

Explanation

A Random Forest Classifier was trained using the training dataset. This algorithm combines multiple decision trees to improve prediction accuracy.

Prediction Results

```
[8]: y_pred = model.predict(X_test)
```

```
print(y_pred[:10])
```

```
[0 0 1 1 0 1 1 0 1 1]
```

Figure 4: Sample Predictions

Explanation

The trained model generated survival predictions for unseen test data.

Model Accuracy

```
[9]: from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)

Model Accuracy: 0.7877094972067039
```

Figure 5: Accuracy Score

Explanation

The model achieved an accuracy score indicating how effectively it predicted passenger survival outcomes.

Confusion Matrix

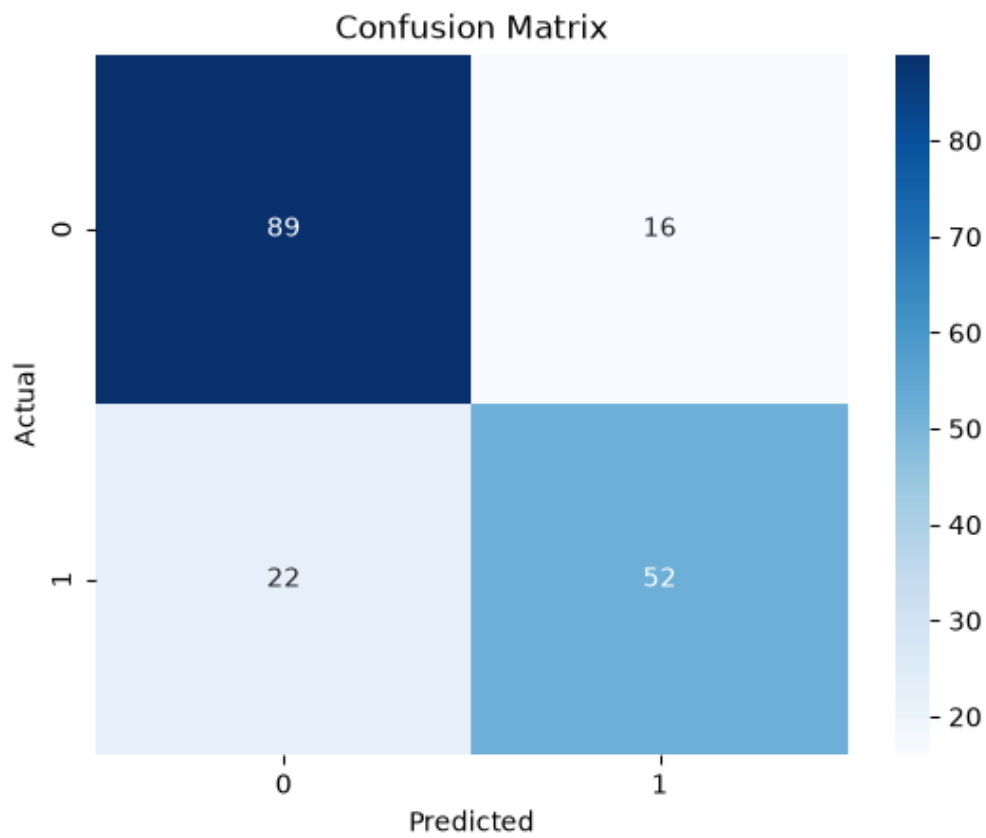


Figure 6: Confusion Matrix

Explanation

The confusion matrix provides a detailed breakdown of correct and incorrect predictions made by the machine learning model.

Conclusion

Key Findings

- Data was successfully preprocessed.
- A Random Forest model was trained.
- Survival predictions were generated successfully.
- The model achieved good prediction accuracy.

Conclusion

This project demonstrated the application of machine learning techniques for predictive analysis. The Random Forest algorithm successfully predicted passenger survival using the Titanic dataset, providing practical experience in model training, testing, and evaluation.